Dzirive, Vanga ward, Kwale County, Kenya

Here El Niño arrives as Umba River floods and the disease that follows them, not as heat. The single most important action is to get treatment stock inside Vanga before the road is cut at Jego and Jasini.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Not retrievable at this resolution. Containing geography: Kwale County (~867,000, 2019 census)	No figure published for Kenya or East Africa	n/a (global: 451,000, Jun 2026 – Feb 2027)	Riverine flood, cholera, and malaria

10/11 recommendations verified. 1 confirmed, 9 corrected, 1 unverified (not sent to a verifier), 0 contraindicated. **The funding target moved 86% in verification**, from ~$189,900 to ~$25,800. Almost all of that is a $147,000 net purchase that is only needed if Kwale is left out of the November national distribution. Exposure figure: Climate Impact Lab, September 2026, pre-peer-review. It covers *heat only* and publishes no Kenyan figure, so no excess-death number appears here, scaled or otherwise. **Heat is not the story on this coast.** The local record is carried by the NDMA and press sources below.

EXECUTIVE SUMMARY — AT A GLANCE



4.65°S 39.20°E, Kenya · Natural Earth

- **The risk.** Umba River floods from Tanzanian rain cut the Vanga road at Jego and Jasini; in the 2023 short rains Kwale had 13 flood deaths (7 in Lunga Lunga), malaria rose 40%, and cholera followed in 2025.
- **First action.** Kwale County Health and the Lunga Lunga sub-county health team, with Kenya Red Cross, put ORS, zinc, IV fluids, cholera kits, malaria tests, and ACTs inside Vanga by 15 October 2026.
- **The money.** ~$18.3k floor, ~$25.8k target, ~$175.2k ceiling. The top placement is Kenya Red Cross early cash to 441 Vanga households on a Coast severe-weather advisory, $23,373.
- **Do not.** Do not post officers at an overtopped bridge and call it closed: at Ramisi on 17 Nov 2023 two KRA officers talked their way past police and drowned.*

What to do	**11** actions	p. 2
Who does what	RACI chart	p. 12
Where money does the most	funding	p. 13
Do not do this	**4** items	p. 14
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How this was produced	methodology	p. 19
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* Suggested formulation: check with a health worker before sharing.

- **Where.** Dzirive is in Vanga ward: 44,101 people in 8,828 households (2019 census, via NDMA). The ward sits on the Umba River delta at the Tanzania border, and people live by fishing and rice farming. Vanga has 980 registered fishers, of whom only 450 are licensed. The rice blocks behind the failed 2019 Umba works now lie fallow from saltwater intrusion. No published record names Dzirive itself.
- **The water comes from Tanzania.** The Umba rises in the Usambara mountains. Kwale's county met director (2022) and Governor Achani (3 May 2024) both said Vanga, Kiwegu, Jego, and Majoreni flood when it rains *there*, so the river can rise on a dry day in Vanga. A 2013 Kenya–Tanzania framework lists data sharing on the Umba, but no working gauge, flood warning, or GloFAS forecast point was found.
- **The road is the hazard.** In the 2023 short rains the Lunga Lunga–Vanga road was impassable at Jasini and Jego (NDMA). In a separate flood, undated in the source, Vanga was cut off to vehicles for four days. The A14 is the only road to Msambweni hospital and Mombasa. The documented deaths are drownings at crossings: two KRA officers at Ramisi bridge on 17 Nov 2023. Kwale recorded 13 flood deaths that season, 7 of them in Lunga Lunga sub-county.
- **Disease follows the water.** Malaria cases in Kwale rose 40% in July–December 2023 over the same months of 2022, which NDMA put down to low net use. That season recorded no cholera. In June 2025 cholera crossed the border into Vanga and Jasini: 48 suspected cases and 3 deaths, with a Lunga Lunga attack rate of 18.1 per 100,000. Kwale is also a historically enzootic Rift Valley fever district.
- **The county averages hide the risk.** In Kwale (KDHS 2022):
  - 47% of households have basic drinking water and 32% basic sanitation.
  - Child stunting is 23%, against 18% nationally.
  - 62% of people slept under a bednet.
  - Under-5 mortality is 30 per 1,000, below Kenya's 41.

  So county figures will not flag Vanga. Deaths here come from delay: a closed road, or an infection that arrives across the border.
- **Heat is not the story.** Climate Impact Lab's projection covers heat only and publishes nothing for Kenya or East Africa, so this briefing carries no excess-death figure. Heat was not assessed by any lens and is listed as a blind spot.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the short rains, which begin around mid-October. Backing lenses are shown per action. Convergence was checked for shared bias: the ORS and bednet evidence is canonical global literature, so it is ranked on the local record (road

* Suggested formulation: check with a health worker before sharing.

cuts, cholera imported across the border in 2025, and malaria up 40% in 2023), not on lens agreement.

1. Put ORS and zinc, IV fluids, cholera kits, malaria rapid tests, and ACTs inside Vanga by 15 October, before the road floods

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · HISTORIAN · SKEPTIC · ECONOMIST · PHILANTHROPIST

ADAPTATION INVENTORY: RELATED PRECEDENT, UGANDA COMMUNITY HEALTH WORKERS CUT WEATHER-SHOCK INFANT DEATHS (2023)

OWNER	BY WHEN	COST	PREVENTS
Kwale County Health / Lunga Lunga sub-county health team, with Kenya Red Cross	15 Oct 2026	Not costed. Payer: Ministry of Health El Niño plan (KES 2.9B, *draft, not approved*)	Diarrhoea and malaria deaths while Vanga is cut off

Where to stock: Vanga health centre, Jasini, and community health promoters' homes.

Why it matters here: in the 2023 short rains the Lunga Lunga–Vanga road was impassable at Jasini and Jego. Malaria cases in Kwale rose 40% (33,515 to 46,948, Jul–Dec 2022 vs 2023), and the county recorded 13 flood deaths, 7 of them in Lunga Lunga (NDMA). In June 2025 cholera crossed the Tanzania border into Vanga and Jasini: 48 suspected cases and 3 deaths, with a Lunga Lunga attack rate of 18.1 per 100,000.

Evidence it works: ORS was associated with 69% lower diarrhoea mortality in under-5s (95% CI 51–80; Munos et al. 2010).

Correction: the 69% pools only three quasi-experimental studies, not trials, and covers ORS alone; zinc has separate evidence.

* Suggested formulation: check with a health worker before sharing.

Transferability: reasonable. It is low-income child diarrhoea evidence, and the local binding constraint is physical access, which pre-positioning addresses directly.

2. Agree before the rains who issues warnings, and on what trigger, then send one message through chiefs, imams, Radio Kaya, and the Vanga fishing co-op (BMU)

CONFIRMED | SKEPTIC · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND (CLOSEST IS US FORECAST ACCURACY, SHRADER 2023)

OWNER	BY WHEN	COST	PREVENTS
Kwale Disaster Management Unit + county met director + Lunga Lunga Deputy County Commissioner	10 Oct 2026	Near zero. Unfunded: no line exists, and none is needed	People ignoring a correct forecast because officials contradict each other

On 22 Oct 2023 President Ruto said there would be "no El Niño." On 10 Nov 2023 the Kenya Met director apologised for the "miscommunication" and confirmed El Niño was under way. The Skeptic's point is that the worst documented failure was not a bad intervention; it was officials contradicting each other.

Channels: Friday sermons, madrassas, a Digo-language segment on Radio Kaya, and the Vanga BMU. Unlicensed fishers (about 530 of Vanga's 980 registered fishers are unlicensed) avoid anyone in uniform, so warnings to them should come from peers at the landing site.

* Suggested formulation: check with a health worker before sharing.

Scope and caveats: the 2023 contradiction was national, not in Kwale. Radio Kaya's frequency and languages are self-reported and should be confirmed locally.

3. Physically close overtopped crossings with barriers and a no-override order, keep rescue crews on standby, and keep a medical relay open across each closure *

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Deputy County Commissioner Lunga Lunga, National Police, KeNHA (national roads authority), and Coast Guard / Kenya Red Cross	Standing order by 10 Oct 2026	Existing police and roads budgets. Barriers not costed	Drownings at crossings, the documented cause of flood deaths here

Which crossings: the Ramisi and Perani bridges on the Likoni–Lunga Lunga highway, Marenje bridge, and the Umba crossings.

The 2023 case: on 17 Nov 2023 two KRA officers were swept off the overtopped Ramisi bridge. Police posted there had ordered them to stop, and they talked their way past. Barrier roadblocks went up only afterwards (The Star). A boda rider's body was recovered in the same search (Nation). The day before, the Coast Guard rescued 26 people at Ramisi, including 18 KCSE candidates, with no deaths.

Corrections: the April 2026 report of Ramisi and Perani being impassable mentions no missing rider. Marenje was partly, not fully, washed away in 2020.

Evidence and transferability: the evidence that vehicle drownings dominate flood deaths is from the US and Europe (45–63% of deaths), and no study shows barriers reduce deaths. Ramisi is the local argument.

The added requirement: the A14 is the only road to Msambweni hospital and Mombasa, so every closure needs a boat relay or an ambulance waiting on each bank. Without one, closing the road just moves the deaths to the referral queue.

* Suggested formulation: check with a health worker before sharing.

Authority: KeNHA can close roads under the Roads Act 2007. The Coast Guard's Act covers rivers, but whether it has rescue assets in Kwale is unverified.

4. Watch rainfall in Tanzania, not in Vanga: set up an upstream Umba relay from the Tanzanian side to the Vanga chief

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MGR · HISTORIAN · SKEPTIC · PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Kwale Disaster Management Unit, with the Pangani Basin Water Board (Tanzania) and WRA Kwale	15 Oct 2026	Airtime. Unfunded	Surprise floods on dry local days; late evacuation of Vanga, Jego, Kiwegu, and Majoreni

The Umba rises in Tanzania's Usambara mountains. Kwale's county met director (2022) and Governor Achani (3 May 2024) both said Vanga floods when it rains in Tanzania.

A 2013 Kenya–Tanzania Joint Cooperative Framework (covering Chala, Jipe, and the Umba) lists data sharing, but **no operational warning, public gauge, or GloFAS point on the Umba was found**. The first step is two phone calls: to the Pangani Basin Water Board and WRA Kwale, asking whether any Umba gauge exists.

* Suggested formulation: check with a health worker before sharing.

Evidence: there is no measured mortality effect for community flood warnings; the literature is descriptive. This ranks high on the local mechanism, not on an effect size.

5. Kenya Red Cross headquarters requests an IFRC "imminent event" DREF allocation for the south coast, to pay for pre-stocking and early cash

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST

ADAPTATION INVENTORY: PRECEDENT, ANTICIPATORY CASH FOR FLOODS IN BANGLADESH 2021 AND 2025 AND NEPAL 2025

OWNER	BY WHEN	COST	PREVENTS
Kenya Red Cross HQ (the request must come from HQ, not the Kwale branch)	Request filed on the first KMD severe-weather advisory for the Coast	Reference: the 2023 Tana activation was CHF 192,698 for 150,000 people	Money arriving only after the photographs, not before the flood

What exists: Kenya Red Cross's riverine-flood Early Action Protocol covers only the Tana, Nzoia, and Athi rivers. **The Umba is not covered.** The IFRC imminent-event DREF, available to a Red Cross society with no protocol for the hazard, can fund readiness and pre-stocking for up to six months. No 2026 El Niño allocation for Kenya has been found yet.

Correction: the 2023 Tana activation was triggered by observed river levels under an IFRC exception; the GloFAS forecast triggers "were never realized." The longer-term fix is adding an Umba trigger to the protocol, which will not happen before this peak.

6. Before opening any evacuation site, check it is above the flood line and stock it with water treatment, latrine kits, and food

PARTIALLY CONFIRMED | SKEPTIC · EMERGENCY MGR · HISTORIAN

ADAPTATION INVENTORY: PRECEDENT, WASH AND SHELTER PREPAREDNESS IN SINDH, PAKISTAN (AVDEENKO & FRÖLICH 2025)

OWNER	BY WHEN	COST	PREVENTS
Ward administrator and chiefs, with Kenya Red Cross Kwale	20 Oct 2026	Not costed; fundable under action 5	Camps turning into cholera sites

* Suggested formulation: check with a health worker before sharing.

Comparable cases: Tana River had 57 cholera cases across 88 camps in May 2024. In Garissa in 2023, MSF's latrine response began weeks after the camps opened. **The local warning:** Kiwegu, the 2023 camp site, itself flooded in April 2026, with more than 200 houses collapsed, so the named halls, mosques, and churches must be checked against the flood line before they are designated.

Transferability: the Pakistan RCT measured food security and health, not deaths, and was a multi-year NGO programme. That is weak support for a two-week push; the local case rests on the camp-cholera pattern.

7. Confirm with the National Malaria Programme that Kwale is in the November 10-million-net distribution, and get existing nets hung now

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Kwale county health management team (CHMT); community health promoters hang the nets	Phone call this week; nets hung by mid-Oct	Zero to confirm. Top-up up to $132,306 only if Kwale is excluded	A repeat of the 2023 malaria surge, which NDMA put down to low net use

Evidence: ITNs cut all-cause child mortality by 17% (RR 0.83, 0.77–0.89), or 5.6 lives per 1,000 children per year (Cochrane, high certainty). Kenya was one of the trial sites. **Local need:** 62% of Kwale residents slept under a net in 2022, and use tracks access (63%), so the gap is nets, not behaviour. The May 2024 nets will be about 2.5 years old at the peak.

* Suggested formulation: check with a health worker before sharing.

Caveats: the trials ran from 1987 to 2001, before pyrethroid resistance. Nets arriving "from November" land after the rains start, and a county-bought top-up probably cannot be procured before the peak.

8. Vaccinate livestock against Rift Valley fever before the rains if the Veterinary Services directorate includes Kwale, using Clone 13 or keeping Smithburn away from pregnant animals

PARTIALLY CONFIRMED | HISTORIAN (PUBLIC HEALTH DISSENTED; ITS BASIS WAS WRONG) | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Kenya's Directorate of Veterinary Services (DVS), with Kwale County Veterinary Department	Before mid-Oct rains	National vaccine stock (DVS says it is pre-positioned); county delivery not costed	An RVF epizootic that spills over to people

Kwale is a historically enzootic RVF district, with activity in 1961–64 and 2006–07 (Murithi et al. 2011). In 2006–07, vaccination began on 8 Jan 2007, after cases had peaked. DVS says it is preparing strategic vaccination for 2026 but has published no county list.

Correction: the Public Health lens's claim that Kwale was outside the 2006–07 outbreak came from an entomology paper that studied four districts, not the full case distribution. **Safety:** Smithburn vaccine causes abortions and birth defects in pregnant animals; Clone 13 does not. Alongside vaccination, tell herders not to handle aborted or dead animals.

9. Distribute household chlorine only with a demonstration and follow-up checks, not as a hand-out *

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC · EMERGENCY MGR | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Community health promoters, supplied by County Health / Kenya Red Cross	Mid-Oct, with the pre-positioned stock	Not costed	Diarrhoea from flood-contaminated water (47% of Kwale households have basic water access)

Evidence: diarrhoea RR 0.77 (0.65–0.91) across 14 trials, low certainty, with high heterogeneity (Clasen 2015, Cochrane).

* Suggested formulation: check with a health worker before sharing.

Use is the binding constraint: in emergencies, effective use ranged from 0% to 67.5%, and passed 20% only where households already knew the method (Lantagne & Clasen 2012). In WASH Benefits Kenya, detectable chlorine fell from 40% to 20% of households over two years, with no effect on diarrhoea.

10. Draft the cholera vaccine request to the ICG now, ready to file on the first confirmed Kwale case, instead of a pre-emptive campaign

PARTIALLY CONFIRMED | PUBLIC HEALTH (PROPOSED A PRE-EMPTIVE CAMPAIGN); PHILANTHROPIST (OPPOSED) | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Kwale County Health, through the Ministry of Health, to the ICG (the global emergency vaccine stockpile)	Dossier ready by 15 Oct	Vaccine from the ICG stockpile; delivery not costed	Weeks of paperwork lag once cholera crosses the border

Why not pre-emptive: preventive oral cholera vaccination restarted in February 2026, but only through Gavi-approved multi-year plans. The first doses went to Mozambique, DRC, and Bangladesh, and **Kenya has no approved allocation on record**. As of July 2026, Kenya had cholera only in Garissa and Nairobi.

Evidence: single-dose efficacy is 40% overall, with no significant protection under age 5 (Qadri 2016, Bangladesh).

Correction: the 18.1/100,000 attack rate is from 2025 outbreak reporting, not the GTFCC report that covers 2018–2023. Whether Lunga Lunga is a cholera priority area (PAMI) could not be confirmed.

11. Walk the riverbank households with the community health promoter, and give every isolated person a named helper and a named destination

UNVERIFIED (NOT SENT TO A VERIFIER) | FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Village elders, community health promoters, and volunteers	Mid-Oct 2026	Volunteer time. Unfunded	The elderly, disabled, and female-headed households being left behind

* Suggested formulation: check with a health worker before sharing.

This action is volunteer-only and low-risk, but no independent check of its evidence was run. Treat it as a strong local hypothesis. A community health promoter count for Lunga Lunga was not found.

* Suggested formulation: check with a health worker before sharing.

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Kwale County Health (CHMT)	Ministry of Health	Kenya Red Cross HQ	Kenya Red Cross Kwale	IFRC	Kwale Disaster Mgmt Unit & county met	Deputy County Commissioner	Police, KeNHA & Coast Guard	Pangani Basin Water Board & WRA Kwale	Ward administrator & chiefs	Community health promoters	Village elders & volunteers	Veterinary Services (DVS & county)	National Malaria Programme
1. Pre-stock ORS, fluids, and tests in Vanga	**A/R**	**C**		**R**							**R**			
2. One warning voice, one agreed trigger						**A/R**	**R**			**R**				
3. Close overtopped crossings; keep medical relay	**C**			**R**			**A**	**R**						
4. Upstream Umba relay from Tanzania						**A/R**			**R**	**I**				
5. Request an IFRC imminent-event DREF			**A/R**	**R**	**C**									
6. Evacuation sites above flood line, stocked			**I**	**R**						**A/R**				
7. Confirm Kwale's nets; hang existing nets	**A/R**										**R**			**C**
8. Rift Valley fever livestock vaccination													**A/R**	
9. Household chlorine with demonstration and checks	**A**			**R**							**R**			
10. Cholera vaccine request to ICG, ready	**A/R**	**R**												
11. Riverbank walks; named helper per person											**R**	**A/R**		

What the chart shows. **Kwale County Health is accountable for four rows, but the money behind them is not liquid.** The Ministry of Health El Niño plan is still a draft, and the county Emergency Fund holds about KES 1M. **The Kenya Red Cross Kwale branch carries an R on five rows.** Much of that work is funded through the DREF request in row 5, which only headquarters can file. **Row 4 crosses a border.** The rain that floods Vanga falls in Tanzania, so the warning depends on the Pangani Basin Water Board, which Kwale cannot direct, and no operational warning or public gauge on the Umba was found. **Row 11 has no named public owner in the report.** It rests on village elders, community health promoters, and volunteers, it is unfunded, and it was not sent to a verifier.

* Suggested formulation: check with a health worker before sharing.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
~$18,300	~$25,800	~$175,200	Whether Kwale is in the November national net allocation ($132k swing)

Floor: 300 households × $53 + $2,400 stock = $18,300. The 300 is Kenya Red Cross's count of households *affected* in Lunga Lunga in April 2026. It is used as a proxy for delivery capacity, which remains undemonstrated.

Target: 441 × $53 + $2,400 = $25,773. The 441 is NDMA's count of Vanga households needing *food assistance* in March 2024, not flood need.

Ceiling: 764 × $53 + $2,400 + 22,051 nets × $6 = $175,198. The 764 is Lunga Lunga's 2023 flood-affected households across four wards, so it is an upper bound for Vanga; 22,051 nets is one per two people.

How the numbers moved: the target fell 86% from the lens's $189,900. The net purchase came out of the target. The lens's $617k ceiling counted all 8,828 households, 29× the delivery capacity actually observed.

Status of the inputs: $53 is a Bangladesh transfer value (WFP 2020), not a Kenyan price. Kwale's county Emergency Fund holds about KES 1M, and FLLoCA climate money cannot be reprogrammed before December, so the public money that could fill this is not liquid before the peak.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Kenya Red Cross (HQ, delivered by Kwale branch via M-Pesa)	Action 5: early cash to Vanga households on a Coast severe-weather advisory	$23,373 (441 × $53)	Anticipatory cash transfer, ideally matched by an IFRC imminent-event DREF	WFP Bangladesh 2020: 36% (WFP) to 52% (Oxford CSAE) fewer days without food. Welfare, not deaths. The "12% more evacuation" figure was not found	PARTIALLY CONFIRMED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Lunga Lunga sub-county health management team	Action 1: ORS, zinc, rapid tests, and ACTs inside Vanga	~$2,400 (placeholder)	Pre-positioned stock	The lens cited a KES 309,184 county zinc line; it was not found in any Kwale budget. Treat as a floor, not a costing	UNVERIFIED
Kwale County Dept of Health, via community health promoters	Action 7: net top-up, *only if* Kwale is excluded nationally	Up to $132,306 (ceiling only)	Pre-positioned stock	GiveWell: ~$6/net, $3,000–8,000 per life saved (AMF, multi-country Africa)	UNVERIFIED, CONTINGENT

The funder's one thing: the cheapest life-saving money here is a phone call. Confirm Kwale's share of the 10 million nets before disbursing anything. If Kwale is covered, the right amount for Vanga is about $26k, not $190k. The second call is to Kenya Red Cross headquarters about an imminent-event DREF, which would put IFRC money, not a private grant, behind action 5.

DO NOT DO THIS — CONTRAINDICATED

No recommendation received a CONTRAINDICATED verdict. The practices below are not panel recommendations. They are the ways local responses have actually gone wrong, each with a documented case.

Posting officers at an overtopped bridge and calling it closed

DOCUMENTED DEATHS

Ramisi, 17 Nov 2023: police ordered two KRA officers to stop. They talked their way past and drowned. Barriers went up only afterwards.

* Suggested formulation: check with a health worker before sharing.

Instead: physical barriers, a no-override standing order, and a medical relay across the gap (action 3).*

Using Smithburn RVF vaccine on pregnant livestock

`CONTRAINDICATED FOR PREGNANT ANIMALS`

The live Smithburn strain causes abortions and birth defects in pregnant animals. Herders who lose calves to a vaccine stop presenting animals for the next campaign.

Instead: Clone 13, or exclude pregnant animals (action 8).

Opening evacuation camps first and adding water and latrines later

`DOCUMENTED HARM, COMPARABLE SETTINGS`

Tana River, May 2024: 57 cholera cases across 88 camps. Garissa, 2023: MSF's latrine build began weeks after the camps opened. On this coast Kiwegu, the 2023 camp site, itself flooded in April 2026.

Instead: check each site against the flood line and stock it with water treatment and latrines before opening (action 6).

Starting RVF vaccination once human cases appear

`DOCUMENTED FAILURE, KENYA 2006–07`

Human cases peaked on 24 Dec 2006, and livestock vaccination began on 8 Jan 2007. Vaccinating during an outbreak does not help.

Instead: vaccinate before the rains, on the forecast (action 8).

* Suggested formulation: check with a health worker before sharing.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1961–64	RVF activity in Kwale (Murithi et al. 2011)	Not recorded	Kwale classed as an enzootic RVF district
1997–98	El Niño floods; FAO named the Umba riverbanks as worst hit	No Kwale toll found	Nothing documented locally
Nov 2006	Floods: ~60,000 affected in Kwale District; 1,000 marooned in Vanga, which was cut off (Kenya Red Cross, seen in search excerpts only)	Not found	Nothing documented
2006–07	El Niño RVF epidemic (coast cases in Kilifi, Tana River, and Malindi)	National; no Kwale figure	Lesson recorded: vaccination came after the peak
2019	Floods swept away homes and livestock in Vanga and Mwangwei	Not found	KES 379M allocated to rehabilitate the Umba. Residents say the work failed and the diversion wall collapsed
2020–21	Marenje bridge partly washed away (2020); Umba flooded 350+ acres (2021)	Not found	—
Oct–Dec 2023	El Niño plus positive Indian Ocean Dipole short rains; road cut at Jasini and Jego; Ramisi drownings	**13 dead in Kwale, 7 in Lunga Lunga**; 764 households affected in Lunga Lunga; malaria +40%; zero cholera (NDMA)	Barriers at Ramisi, after the deaths
Jun 2025	Cholera imported across the Tanzania border into Vanga and Jasini	48 suspected cases, 3 deaths (Kwale)	—
Apr 2026	Long-rains floods: Kiwegu, Ramisi, and Perani	200+ houses collapsed	Residents: "a problem we face almost every year"

* Suggested formulation: check with a health worker before sharing.

The recurrence nobody has noticed: every flood in this record ends with the same road cut, at Jego and Jasini, and the water comes from rain in Tanzania, not from rain in Vanga. The Pacific forecast matters less here than the Usambara mountains and the Indian Ocean Dipole: 2023 was a double event. Underneath, the failed Umba works and saltwater on the rice blocks mean each flood lands on households already weakened by the 2020–23 drought.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Run a pre-emptive cholera vaccine campaign** (Public Health) versus **don't fund one; Kwale had zero cases in 2023** (Philanthropist). Feasibility, not argument, settled this: Kenya has no preventive allocation. A pre-filed ICG request (action 10) is the workable middle, though Lunga Lunga's 2025 attack rate shows the Philanthropist's "zero cases" is a single-year view.
- **RVF vaccination is last priority because Kwale was outside the 2006–07 outbreak** (Public Health) versus **vaccinate before the rains** (Historian). Verification sided with the Historian: Kwale is historically enzootic. What would settle it is DVS's 2026 county list.
- **Nets are the best buy per dollar** (Philanthropist) versus **nets bought now land after the rains** (verifier). Both are true. The nets decision is a phone call about the national allocation, not a grant.
- **"Kwale is not short of money, it is short of a trigger"** (Economist) versus **the emergency fund holds ~KES 1M and FLLoCA money cannot move before December** (verifier). Kwale is short of both. That is why the IFRC imminent-event DREF, requested by Kenya Red Cross HQ, is the one realistic before-the-peak instrument.
- **Close the crossings** (Emergency Manager) versus **the A14 is the only road to hospital** (verifier). Resolved by pairing the two: every closure needs a medical relay.

WHAT NOBODY COVERED — BLIND SPOT

Obstetric and emergency referral while Vanga is cut off. No lens asked what happens to a woman in obstructed labour, or a child with severe malaria, during the four-plus days Vanga has been unreachable by road. The crossing verifier raised it only as a side effect of closing bridges. A named boat or alternative route to Msambweni, and a list of women due to deliver October to December, could change actions 1 and 3 more than anything else here.

Also uncovered:

- Heat, a secondary hazard on a hot, humid coast.

* Suggested formulation: check with a health worker before sharing.

- The population on the Tanzanian side of the Umba (Mkinga), who share the floodplain and brought cholera in 2025.
- Fishers at sea during Indian Ocean storms.

HOW TO READ THIS METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

* Suggested formulation: check with a health worker before sharing.

This run. One location, run 28 September 2026. Seven lens agents, then six verifiers checking eleven recommendations and the funding envelope; the verdict tags carry the result. Action 11 (the vulnerability walk) was not sent to a verifier and is marked unverified. The Adaptation Inventory was queried from the full 96-entry dataset embedded in the inventory page, filtered for Kenya, East Africa, flood, and health.

Resolution. Dzirive is a village in Vanga ward (44,101 people, 8,828 households, 2019 census via NDMA). No published record names Dzirive itself, so every figure is for Vanga ward, Lunga Lunga sub-county, or Kwale County, and each says which. Lunga Lunga covers four wards, so its figures overstate Vanga alone. The Climate Impact Lab impact region could not be retrieved at this resolution.

The El Niño projection covers *heat only* and publishes nothing for Kenya or East Africa, so this briefing carries no excess-death figure, scaled or otherwise. Flood, disease, and crop failure are carried by the local record. The Adaptation Inventory holds no entries on cholera, bednets, Rift Valley fever, or river crossings; its relevant precedents are anticipatory flood cash (Bangladesh, Nepal), community health workers (Uganda), WASH preparedness (Pakistan), and index-based livestock insurance (northern Kenya).

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Dzirive, Vanga ward, Kwale County, Kenya. Climate Impact Lab impact region, as the report records it: Not retrievable at this resolution. Containing geography: Kwale County (~867,000, 2019 census). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.

* Suggested formulation: check with a health worker before sharing.

- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (flooded crossings and household water treatment) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (✓ = fetched and checked by a verifier; ◦ = cited by a lens, not independently re-checked)

✓ NDMA Kwale Short Rains Assessment 2023 · knowledgeweb.ndma.go.ke

✓ Munos, Walker & Black 2010, Int J Epidemiol · academic.oup.com

✓ Pryce et al. 2018, Cochrane ITNs · PMC6418392

✓ KDHS 2022 Kwale factsheet · dhsprogram.com

✓ Kwale cholera 2025 · peopledaily.digital

✓ GTFCC PAMI report, Kenya · gtfcc.org

✓ Preventive OCV resumes, WHO, 4 Feb 2026 · who.int · WHO AFRO week 28 bulletin · afro.who.int

✓ Murithi et al. 2011, RVF in Kenya · cambridge.org · CDC MMWR 2007 · cdc.gov · DVS 2026 readiness · the-star.co.ke · RVF vaccine safety · PMC10625542

✓ Clasen 2015 Cochrane · europepmc.org · Lantagne & Clasen 2012 · pubmed · WASH Benefits Kenya · thelancet.com

✓ Ramisi drownings · the-star.co.ke · nation.africa · Coast Guard rescue · the-star.co.ke · April 2026 floods · the-star.co.ke

✓ Jonkman & Kelman 2005 · wiley.com · Kenya Coast Guard Service Act · kenyalaw.org

✓ Governor Achani, 3 May 2024 · kenyanews.go.ke · County met director 2022 · the-star.co.ke · Kenya–Tanzania 2013 MoU · iisd.org

* Suggested formulation: check with a health worker before sharing.

✓ Ruto "no El Niño" · capitalfm.co.ke · KMD apology · the-star.co.ke

✓ MSF Garissa · msf.or.ke · Tana River cholera 2024 · reliefweb.int · Avdeenko & Frölich 2025 · repec

✓ KRCS riverine-flood EAP · anticipation-hub.org · EAP2021KE01 · IFRC Anticipatory Pillar · ifrc.org

✓ MoH El Niño plan (draft, KES 2.9B) · nation.africa · 10M nets · allafrica.com

✓ FLLoCA disbursement · climate.co.ke · Kwale FY2025/26 budget · kwale.go.ke · World Bank RRO · kenyans.co.ke · FAO–WFP appeal · wfp.org

✓ GiveWell top charities · givewell.org · WFP Bangladesh anticipatory cash · csae.ox.ac.uk · anticipation-hub.org

◦ FAO/GIEWS 1997–98 · fao.org · Umba works and salinisation · willowhealthmedia.org · Vanga fishers · kenyanews.go.ke · Radio Kaya · radiokaya.co.ke · Vanga cut off · nation.africa

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 28 September 2026 for Dzirive, Vanga ward, Kwale County, Kenya. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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